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The Role of Nutrition in Endometriosis Prevention and Management: A Comprehensive Review

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Abstract

Endometriosis is a chronic gynecological condition defined by the ectopic presence of endometrial-like tissue outside the uterine cavity, often resulting in debilitating symptoms and significant impacts on quality of life. While the exact etiology of endometriosis remains elusive, emerging evidence suggests that diet and nutrition may play a crucial role in its pathogenesis and management. This comprehensive review explores the complex interplay between various food substances and endometriosis, summarizing the latest research findings on both risk-enhancing and protective nutritional factors. Notably, consumption of alcohol, red and processed meats, foods high in saturated and trans fats, and excessive caffeine has been correlated with increased systemic inflammation and hormonal dysregulation-key mechanisms implicated in endometriosis pathogenesis. In contrast, nutrients such as antioxidants, B-complex vitamins, vitamin D, calcium, polyunsaturated fatty acids (PUFAs, particularly omega-3 and omega-6), and dietary fiber have shown promise in exerting anti-inflammatory and protective effects against endometriosis. The review emphasizes the importance of promoting a balanced and nutritious diet rich in anti-inflammatory and antioxidant-rich food while limiting the intake of pro-inflammatory substances for individuals with endometriosis. In addition to dietary interventions, lifestyle modifications, including regular physical activity, stress management, and optimizing sleep hygiene, are highlighted as integral components of comprehensive treatment plans for endometriosis patients. Further research is required to clarify the precise mechanisms underlying the relationship between diet and endometriosis and to establish evidence-based dietary recommendations personalized for patients with endometriosis.

Keywords: Diet, Endometriosis, Inflammatory Factors, Nutrition, Protective Factors

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Introduction

Endometriosis is characterized as a chronic, inflammatory gynecological disease that affects approximately 10% of women of reproductive age worldwide. Despite advancements in diagnostic and therapeutic modalities, the global incidence of endometriosis continues to rise. This trend is likely underestimated, particularly in low- and middle-income countries, due to pervasive underdiagnosis and undertreatment (1). Endometriosis refers to the growth of endometrium-like tissue outside the uterus, which results in debilitating symptoms including dysmenorrhea, dyspareunia, dyschezia, dysuria, infertility, and chronic pelvic pain. Moreover, a significant proportion of patients with endometriosis also reported depression and anxiety, as well as a lower quality of life (2).

Recent studies revealed that diet and nutrition may significantly influence the progression of endometriosis. Several dietary components have been associated with the exacerbation of the disease, while others appear to offer protective effects (3, 4). Growing evidence suggests that consuming the right diet may play a significant role in managing endometriosis. Encouragingly, a survey by Mazza et al. (5)

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found that as many as 64.4% of women with endometriosis decided to change their dietary composition and habits after receiving the diagnosis. This shows a high interest from the general public in the potential role of nutrition in health and diseases, particularly in relation to endometriosis.

Despite growing interest, the precise influence of specific macronutrients and micronutrients on the pathogenesis of endometriosis remains inadequately defined. Moreover, it is still uncertain whether dietary modifications can be sufficiently effective to serve as a standalone therapeutic intervention. One such dietary strategy is referred to as the orthomolecular approach, which involves the therapeutic use of naturally occurring substances-nutrients endogenous to the human body-to restore optimal physiological function. This review aims to comprehensively summarize the most recent updates on the relationship between various food substances and endometriosis in a comprehensive manner.

This literature review aims to synthesize current research findings on the association between dietary factors and endometriosis. The literature search was conducted using multiple electronic databases, including PubMed, Google Scholar, and ScienceDirect, supplemented by manual searches of relevant journals. Targeted keywords used during the search process included "endometriosis," "diet," "food," "nutrition," "inflammatory factors," "protective factors," and "dietary recommendations." The search focused on peer-reviewed studies published between 2000 and 2024. The inclusion criteria were: i. Studies investigated the association between endometriosis and nutrition, ii. Both primary data from original research and secondary data from past reviews or meta-analysis were included, iii. Studies employing either qualitative or quantitative methodologies. Articles that were not written in English and did not provide full texts were excluded.

The pathogenesis of endometriosis

Endometriosis is widely recognised as an estrogen-dependent condition with several hypotheses proposed to explain its etiology. The most established theories include retrograde menstruation, coelomic metaplasia theory, tissue injury and repair theory, and hematogenous or lymphogenous spread (6). The pathogenesis of endometriosis involves a complex interplay of several mechanisms. According to Sampson's accepted theory, menstrual blood containing endometrial cells regurgitates into the peritoneal cavity via the fallopian tubes, leading to implantation of these cells (7). Risk factors such as shortened menstrual cycles and obstructed uterine outflow may increase the quantity of retrogradely flushed cells, contributing to endometriosis development (6, 7). In addition to mechanical and hormonal factors, immune dysregulation plays a pivotal role in endometriosis, with various immune cells implicated in lesion formation (8). Macrophages, neutrophils, natural killer cells (NK cells), dendritic cells, and T cells contribute to inflammation and lesion progression through altered function and cytokine release (8). A previous study found that there is inhibition of immune response in the endometriosis microenvironment, particularly through suppression of NK cell activity and T-cell response (9). Hormonal imbalance, epigenetic modifications, and environmental exposures such as dietary factors and pollutants are also believed to modulate susceptibility to endometriosis (10).

In summary, understanding the multifaceted mechanisms involved in endometriosis pathogenesis is crucial for developing effective diagnostic and therapeutic strategies for this condition. Key contributors include retrograde menstruation, immune dysregulation, hormonal imbalance, genetic and epigenetic factors, stem cell dynamics, and environmental influences (6, 11). These diverse and interrelated factors underscore the complexity of endometriosis and highlight the necessity for integrated, multidisciplinary approaches in its clinical management and future research.

Current diagnostic and treatment modalities

Current diagnostic and therapeutic approaches for endometriosis present significant challenges due to symptom variability and the absence of a universally accepted non-invasive diagnostic tool. Although clinical assessment and pelvic examination may suggest endometriosis, the presence of asymptomatic cases and the poor correlation between symptom severity and disease extent complicate accurate diagnosis. While pelvic examination abnormalities often correlate with endometriosis, laparoscopic confirmation reveals the disease in over 50% of women with normal pelvic exams, highlighting limitations in relying solely on physical examination (12, 13). Conventional diagnostic modalities, including imaging techniques and blood-based biomarkers, offer limited diagnostic sensitivity and specificity. However, emerging research on microRNAs (miRNAs) shows promise as potential non-invasive diagnostic biomarkers. Dysregulated expression patterns of specific miRNAs have been identified in endometriosis, suggesting potential for adjunctive tools in the diagnostic evaluation. However, further validation and standardization are essential to fully exploit the potential of miRNAs in improving diagnostic accuracy for endometriosis (14).

Recent advancements in analytical methodologies and artificial intelligence have significantly contributed to the identification of promising biomarkers and therapeutic targets for endometriosis management. Machine techniques, such as support vector machine (SVM) algorithms, have been employed to construct predictive diagnostic models, identifying five key biomarkers with potential clinical relevance. Concurrently, molecular investigations utilizing single-gene gene set enrichment analysis (GSEA) have elucidated critical pathways involved in the pathogenesis of endometriosis. The development of a competing endogenous RNA (ceRNA) regulatory network further elucidates the complex interactions among these biomarkers. Potential therapeutic drugs targeting these biomarkers are also highlighted, offering novel avenues for endometriosis treatment. Clinical validation of biomarker expression reinforces the significance of these findings in advancing both the diagnosis and treatment of endometriosis (15). Minimally invasive surgery, particu-

larly laparoscopy, offers substantial advantages, including reduced rates of surgical complications and shorter hospital stays (13). While robotic-assisted laparoscopy yields comparable perioperative outcomes, it may provide additional advantages in advanced-stage (stage III and IV) endometriosis cases requiring complex and extensive resection. Certain aspects of endometriosis surgery, such as lesion heterogeneity and difficulty predicting surgical complexity, may benefit from robotic technology, emphasizing its potential to enhance patient care in challenging cases (16).

Dietary factors linked to aggravating endometriosis

Alcohol

Alcohol consumption has long been associated with various chronic inflammatory conditions. In relation to endometriosis, alcohol may contribute to elevated circulating estrogen levels in the blood through increasing aromatase activity, an enzyme responsible for converting testosterone to estrogen. Frydenberg et al. (17) reported that women consuming more than 10 g of alcohol per day exhibited an 18% higher mean 17β-estradiol level compared with those consuming less than 10 g per day. These findings suggest that alcohol consumption affects the cumulative estrogen level in the body throughout a woman's life cycle (4, 17, 18).

A recent systematic review and meta-analysis conducted by Li Piani et al. (19) found that there was a significant association between moderate alcohol intake and endometriosis [unadjusted odds ratio (OR): 1.22, 95% confidence interval (CI): 1.03-1.45, P=0.02].

The metabolism of alcohol also upregulates pro-inflammatory and oxidative stress-related pathways, which further lead to the production of persistent inflammatory mediators and free radicals (4). Excessive reactive oxygen species (ROS) lead to upregulation of transcription factor nuclear factor kappa B (NFkB), which is implicated in the pathogenesis of endometriosis. Following this transcription factor regulation, activated peritoneal macrophages will express many genes associated with proinflammatory cytokines, growth factors, angiogenic factors, chemokines, and others. This chain of events eventually results in endometrial fragment implantation, proliferation, and neovascularization (20-22).

Another concerning issue is how alcohol is often viewed as a coping mechanism by populations affected by chronic conditions, including endometriosis patients (19). Gao et al. (23) reported that individuals with endometriosis are at significantly higher risk of developing alcohol or substance use disorders compared to other psychiatric conditions [hazard ratio (HR): 1.93; 95% CI: 1.71-2.18]. Rather than offering pain relief or a sense of liberation, this actually creates a vicious cycle that further aggravates the disease progression. However, considering the many types of alcohol and varying alcohol percentages in a drink, the dose-response relationship between alcohol and endometriosis remains unclear. Further research is warranted to determine a definitive threshold for alcohol intake that may be considered unsafe in this patient population.

Red meat, processed food, trans and saturated fats

Red meat and processed animal products such as bacon or butter all contain high are rich sources of saturated fats. Excessive intake of saturated fats has been associated with higher levels of estradiol and steroid hormones in the blood. In addition, trans fats, which can be naturally found in meat and dairy and in industrially processed foods such as deep-fried items, have been shown to stimulate the production of tumor necrosis factor-alpha (TNF-α), interleukin-6 (IL-6), and C-reactive protein (CRP) (4, 24). These inflammatory mediators are believed to contribute to the chronic inflammatory milieu observed in endometriosis. Despite this theoretical framework, in practice their role in endometriosis progression is not fully clear. A meta-analysis of randomized controlled trials by O'Connor et al. (25) found that red meat consumption, particularly unprocessed beef and pork, did not significantly affect levels of glycemic control and inflammation biomarkers [fasting glucose, insulin, homeostatic model assessment for insulin resistance (HOMA-IR), HbA1c, IL-6, CRP, TNF-α], within a 16-week observation period. The authors suggested that consuming a balanced, healthy, and nutrient-dense diet in addition to weight loss help achieve control of cardiometabolic disease risk factors independent of total red meat intake (25). In addition, red meat intake can increase risk of endometriosis through other mechanisms. Red meat consumption has been linked with lower sex hormone binding globulin (SHBG) and elevated estradiol levels. High iron content in red meat can also amplify oxidative stress and inflammation. How meat is processed may also influence endometriosis risk; red meat grilled at high temperature can undergo pyrolysis and produce polycyclic aromatic hydrocarbon (PAH), a harmful compound that has been associated with increased risk of endometriosis. PAH can dysregulate levels of reproductive hormones such as estradiol, testosterone, follicle-stimulating hormone (FSH), and luteinizing hormone (LH) (26-28).

One of the few studies that has confirmed a significant correlation between red meat intake and endometriosis is the Nurses' Health Study II by Yamamoto et al. (26), who revealed that women who consumed more than two portions of red meat a day were 56% more likely to develop endometriosis compared to women who consumed red meat once a week. Similarly, an earlier Italian study reported a significantly increased in risk of endometriosis among women who regularly consumed ham, beef, and other red meats (29). Although several other studies have also reported associations between red meat and endometriosis, but none of the results were significant (30, 31). Regarding saturated fats consumption, while older studies comparing the risk of acquiring endometriosis from high intake with low intake found no significant correlation (32-34), a recent meta-analysis found that an increased risk for endometriosis was significantly associated with red meat [relative risk (RR): 1.17, 95% CI: 1.08 to 1.26, P<0.001, I^2=82.4%], saturated fats (RR: 1.06, 95% CI: 1.04 to 1.09, P<0.001, I^2=57.3%), and trans fats (RR: 1.12, 95% CI: 1.02 to 1.23, P=0.019, I^2=73.0%) (35). Additionally, one study specifically examining trans fat in-

take found that women in the highest consumption group were 48% more likely to be diagnosed with endometriosis compared to those with the lowest intake (33).

Most dietary guidelines endorsed by the government or clinicians generally restrict the intake of red meat, saturated fats, and trans fats for better health, such as the dietary approaches to stop hypertension (DASH) and the Mediterranean diet. Although there has been no universally accepted dietary guideline for endometriosis patients, it is good practice to switch to healthier alternatives such as white meat, low-fat diet, and food high in polyunsaturated fats. These dietary modifications and their potential implications for endometriosis will be discussed in the following sections.

Caffeine

Caffeine has been implicated in the modulation of estrogen-dependent disorders such as endometriosis, although the underlying mechanisms remain poorly understood. This active substance can increase plasma levels of steroid hormones, sex hormone-binding protein (SHBG), and therefore lower concentration of free testosterone and estradiol. Caffeine is believed to inhibit aromatase activity, which catalyzes the conversion of androgens to estrogens. Alterations of the hormonal balance are hypothesized to promote endometriosis progression (4, 36). A meta-analysis conducted by Chiaffarino et al. (37) found no statistically significant correlation between caffeine intake and risk of endometriosis [total RR: 1.26, 95% CI: 0.95-1.66 for caffeine and 1.13 (95% CI: 0.46-2.76) for coffee consumption] (37). Similarly, a more recent meta-analysis by Kechagias et al. (36) corroborated the findings that caffeine was not significantly associated with endometriosis (RR: 1.12, 95% confidence interval (CI): 0.97-1.28, I^2=70%) (36). However, it suggested that higher intake of caffeine (more than 300 mg/day) could potentially increase the risk of endometriosis (RR: 1.30, 95% CI: 1.04-1.63, I^2=56%) (36).

Interestingly, the impact of caffeine on estrogen levels appears to vary across racial groups. For example, one study reported that caffeine consumption was associated with decreased levels in white women, whereas it led increased estrogen concentration among Asian women. These differences are hypothesised to result from genetic polymorphism and gene-environment interactions which may vary between races (38). This warrants further research for clarification.

Women with endometriosis are often advised to limit caffeine intake, as it may potentially aggravate their symptoms, such as pelvic pain and cramps. Previous studies have documented self-reported symptom improvement among endometriosis patients who reduced their intake of certain foods, including coffee (5, 39). However, there is still limited evidence regarding this relationship between caffeine and endometriosis-associated pain.

Protective nutrients against endometriosis

Antioxidants

Emerging evidence suggests that patients with endometriosis exhibit lower serum concentration of vitamin A, C, and E than the normal population (40, 41). This deficiency is believed to be linked to oxidative stress. Oxidative stress has been shown to be crucial in endometriosis pathogenesis as it promotes ectopic implantation of endometrial cells (42, 43). Vitamins A, C, and E possess antioxidant properties that neutralize ROS and mitigate oxidative damage within metabolic pathways (40). In addition to their antioxidative function, vitamin C and E have exert anti-inflammatory activity, in which they can downregulate the production of inflammatory mediators such as IL-1 and IL-6. These vitamins may also attenuate endometriosis-associated pain by inhibiting cyclooxygenase activity, thereby reducing prostaglandin E2 synthesis (44). A randomized controlled trial conducted by Amini et al. (45) demonstrated that supplementation with vitamin C and E resulted in significant improvements in dysmenorrhea, dyspareunia, and pelvic pain associated with endometriosis after 8 weeks intervention period.

Vitamin A and its metabolites have been shown to positively regulate gut microbiome-derived butyrate, which plays a role in gut barrier maintenance, suppression of inflammation and cytokine storm, mitochondria optimization, melatoninergic pathway activation, and histone deacetylase (HDAC) inhibition. Butyrate is a short-chain fatty acid (SCFA) that is believed to be decreased in endometriosis due to gut dysbiosis. This observation has contributed to the emerging concept of a "gut–uterus axis," which suggests that endometriosis may be influenced by intestinal health and microbial composition (42). Nevertheless, these findings support the idea that endometriosis should be viewed as a systemic disease rather than a local estrogen-mediated uterine disorder. Furthermore, all-trans retinoic acid (ATRA), a bioactive metabolite of vitamin A, has demonstrated potential in inhibiting the progression of endometriosis by decreasing estrogen levels and preventing endometrial tissue cysts' proliferation (46).

Selenium is a trace element with potent antioxidant which can be found in a variety of dietary sources, including seafood, meat, nuts, some fruits and vegetables, and eggs. Its antioxidative capacity enables it to neutralize ROS, which are implicated in the pathogenesis of endometriosis. Beyond its role in oxidative stress mitigation, selenium also exhibits anti-apoptotic and anti-angiogenic effects, which may further contribute to its protective role in endometriosis (3, 47). Similarly, zinc is an essential mineral with known antioxidant effects that appears to be decreased in patients with endometriosis. Dietary sources of zinc include meat, seafood, dairy products, and supplements. Zinc is involved in DNA synthesis for oocyte formation as well as the proliferation and differentiation of reproductive system cells (3, 47). Zinc deficiency has been linked to several reproductive disorders, including polycystic ovarian syndrome (PCOS), abnormal LH and FSH balance, preeclampsia, and endometriosis (48, 49). A study by Singh et al. (50) showed that the follicular fluid of women with endometriosis had increased concentration of ROS and decreased levels of antioxidant molecules, including

selenium and zinc, compared to that of women with tubal infertility (50). Furthermore, clinical studies have reported that the severity of endometriosis is inversely correlated with oral intake of antioxidant nutrients, including vitamin C, vitamin E, selenium, and zinc (47).

B-complex vitamins

Vitamin B-complex has been proposed as a component of both preventive and therapeutic strategies for endometriosis. A case-control study in Iran demonstrated that consuming a diet rich in vitamins B2, B6, B12, and C, as well as calcium and potassium, may reduce the risk of endometriosis (3). Vitamin B6 plays a critical role in activating the pathways to break down estrogen to prevent excessive levels that can lead to endometriosis proliferation. Consuming vitamin B6 and B9 along with antioxidant nutrients may also alter gene expression or DNA methylation related to endometriosis risk (51). Higher intake of vitamin B12 has been inversely associated with lower plasma concentration of homocysteine, which is a thiol-containing amino acid that promotes oxidative stress and inflammation responsible for endometriosis progression (52). A previous study discovered that endometriosis patients had significantly higher levels of homocysteine in the blood and follicular fluid compared to those without endometriosis (53). Overall, consumption of B-complex vitamins such as vitamin B6, B9, and B12 may contribute to reducing endometriosis risk via multiple biochemical and molecular pathways.

One potential strategy to increase vitamin B levels in patients with endometriosis is through the consumption of more probiotics. Probiotics are natural producers of vitamin B; not only that, but they also maintain gut health, increase the absorption of vitamins and minerals, and boost the immune system (47, 54). Moreover, women with endometriosis should augment their animal protein and vegetable intake as these are the primary food sources for vitamin B12 and B9 respectively (55).

Vitamin D and calcium

Vitamin D is widely recognized not only for its role in calcium homeostasis but also as a key modulator of immune system function (56). Vitamin D can be found in dairy products, seafood, and meat, although it is mainly obtained through adequate sunlight exposure. It is one of the many micronutrients that have been largely studied for its role in endometriosis. Evidence from past studies indicated that the endometrial tissue naturally contains vitamin D receptor (VDR), and it was also revealed that ectopic endometrial tissue had higher VDR levels (57, 58). Up to present, very limited data can be found regarding tissue expression levels of VDR in endometriosis patients. Nevertheless, a recent study by Matasariu et al. (59) was able to find that women with endometriosis who did not receive hormonal treatment exhibited VDR overexpression in parallel with reduced serum levels of 25(OH) vitamin D. Regarding vitamin D serum concentration, many studies have demonstrated that patients with endometriosis had significantly lower vitamin D levels compared to controls (60, 61). As a result, vitamin D deficiency in endometriosis has emerged as an increasingly prominent area of investigation. Typically, vitamin D insufficiency is defined as serum levels between 20-30 ng/mL, while deficiency is classified as levels below 20 ng/mL (56).

VDR acts as both a receptor and a transcription factor that mediates many vitamin D-associated metabolic pathways. Through these signaling pathways, vitamin D is able to suppress key processes involved in the pathogenesis of endometriosis, including inflammation, angiogenesis, cellular adhesion, invasion, and proliferation. It also modulates the immune response by influencing CD4-positive T cells, specifically enhancing T helper 2 (Th2)-mediated anti-inflammatory pathways while inhibiting T helper 1 (Th1)-mediated pro-inflammatory responses (62). Additionally, vitamin D acts on variety of immune cells expressing VDR which include B lymphocytes, monocytes, macrophages, and Langerhans cells (56). However, despite the demonstrated beneficial effect of vitamin D therapy in animal and *in vitro* studies, this effect still lacks conclusive evidence from human trials. Hence, further studies are needed to investigate the effect of vitamin D supplementation using various doses and in various subtypes of endometriosis. Although endometriosis alone may not be sufficient to justify the use of vitamin D supplementation should still be strongly recommended in all women diagnosed with vitamin D deficiency (56).

Calcium absorption relies upon adequate vitamin D levels in the blood. It was suggested that calcium's protective role against endometriosis may involve the prevention of retrograde menstruation through smooth muscle contraction. A case-control study by Roshanzadeh et al. (3) found a significant inverse association between total calcium intake and risk of endometriosis. Similarly, another case-control study based in Korea found that patients with endometrioma had a significantly lower calcium intake than the control group (63). Endometrioma refers to cystic lesions commonly found in the ovaries, also known as deep ovarian endometriosis, which shares a similar etiopathogenesis with endometriosis (64).

Polyunsaturated fatty acids

While saturated and trans fats are associated with an increased risk of endometriosis, it is important to note that not all dietary fats exert the same biological effects. PUFAs, abundant in fish, nuts, seeds, and dietary supplements, play a beneficial role in reproductive health. Since the human body's capacity for producing PUFAs is limited, dietary intake is essential. The two primary categories of PUFAs are omega-3 and omega-6 fatty acids, both of which have been extensively studied in relation to endometriosis.

PUFAs have demonstrated anti-proliferative, anti-angiogenic, and anti-inflammatory properties in both *in vitro* and *in vivo* models of endometriosis. These mechanisms contribute to the suppression of lesion development and progression. Furthermore, PUFA supplementation has been associated with symptomatic improvement, particularly in the reduction of dysmenorrhea (4, 47, 65, 66).

Human studies have also demonstrated positive results regarding the role of PUFAs in the prevention of endometriosis. For instance, Savaris et al. (34) reported that women diagnosed with endometriosis had significantly lower omega-3 consumption than women in the control arm. Similarly, Missmer et al. (33) also showed that women with the highest consumption of omega-3 in their study were 22% less likely to develop endometriosis.

A recent study by Akyol et al. (67) compared the effects of vitamin D supplementation versus omega-3 supplementation on endometriosis lesions in a rat model. The finding concluded that omega-3 was significantly more effective than vitamin D in reducing the volume of the lesion. In the rats given omega-3 supplementation, levels of IL-6, TNF-alpha, and vascular endothelial growth factor (VEGF) in the peritoneal fluid were significantly decreased. Whereas in the rats given vitamin D supplementation, only IL-6 was significantly decreased (67).

Dietary fiber

Dietary fiber consists of complex carbohydrate compounds that are resistant to complete digestion in the gastrointestinal tract. It can mainly be found in fruits, vegetables, whole grains, and beans. The protective role of dietary fiber against endometriosis has been suggested to be mediated by its anti-proliferative effect. This anti-proliferative effect is brought by the role of fiber in estrogen metabolism (4). In contrast, diets rich in simple carbohydrates or low-fiber food have a high glycemic index, leading to a spike in glucose in the bloodstream and subsequent rapid insulin release. Since insulin can act as a growth factor that promotes proliferation and inhibits apoptosis of certain cell types, hyperinsulinism can contribute to endometrial lesion growth. In particular, hyperinsulinism also reduces the level of sex-hormone binding globulin in the blood, which in turn increases the number of circulating free estrogen. Overall, hyperestrogenism due to hyperinsulinism may lead to worsening of endometriosis. This state of hyperestrogenism, driven by hyperinsulinemia, may exacerbate the progression of endometriosis (4, 68-70). Previous studies have shown that low fiber intake is associated with other estrogen-dependent disorders which include breast cancer and endometrial cancer (71). Thus, adopting a high-fiber diet may mitigate the effect of unopposed estrogen. In fact, the concentration of estrogen in the blood can be lowered by 10 to 25 percent by consuming less fats and more fiber (24). Additionally, high fiber consumption may also exert anti-inflammatory effects, further contributing to its protective role (72).

A prospective cohort study by Schwartz et al. (72) observed that women who consumed more food with high glycemic index were more likely to receive a diagnosis of endometriosis. Conversely, although an inverse association was noted between higher fruit fiber intake and endometriosis risk, the finding did not reach statistical significance (72). Despite promising *in vitro* and *in vivo* findings, many clinical studies have struggled to establish statistically significant associations (31). Therefore, more research is needed to clarify this matter regarding the role of fiber in endometriosis prevention. A summary of aggravating versus protective dietary factors in endometriosis can be found in Table 1.

Table 1: Dietary factors and nutrients associated with endometriosis

Dietary factors and nutrients	Potential role in endometriosis	Proposed mechanisms (references)
Alcohol	Aggravating	Raises estrogen levels; increases inflammation; increases oxidative stress (4, 17, 18)
Red meat; processed food; deep-fried food		High in saturated and/or trans fats, which raises inflammatory markers; May alter reproductive hormonal balance (4, 24, 27)
Caffeine		Alters hormonal balance; worsens endometriosis symptoms such as pelvic pain and cramps (4, 36)
Vitamin C and E	Protective	Anti-inflammatory and antioxidant properties; improves endometriosis symptoms such as dysmenorrhea, dyspareunia, and pelvic pain (40, 44, 45)
Vitamin A		Antioxidant activity; decreases estrogen levels through all-trans retinoic acid; maintains gut barrier and suppresses inflammation (42, 46)
Selenium		Antioxidant, anti-apoptotic, anti-angiogenic properties (3, 47)
Zinc		Antioxidant, essential for healthy oocyte formation and proliferation and differentiation of reproductive cells (3, 47)
B-complex vitamins		Decreases estrogen levels; decreases homocysteine levels which helps suppress inflammation and oxidative stress (3, 51, 52)
Vitamin D		Suppresses inflammation, angiogenesis, adhesion, invasion, and proliferation of endometriosis tissue; promotes calcium absorption (56, 62)
Calcium		Prevents retrograde menstruation by enabling smooth muscle contraction (3)
Omega-3; omega-6		High in polyunsaturated fats, which have anti-inflammatory, anti-angiogenic, and anti-proliferative activities (4, 47, 65, 66)
Dietary fiber		Decreases estrogen levels and suppresses inflammation (4)

Which diet regimen should be followed?

To date, there is no universally accepted nutritional guideline specifically established for patients with endometriosis. Such consensus may be difficult to formulate since there is evidence that endometriosis is influenced by genetic and epigenetic factors, in which they may differ according to an individual's ethnicity and geographic location (73). The variety of food sources available and the cost of food in each country may also differ from one another. Nonetheless, there remains a critical need for more comprehensive, evidence-based dietary guidelines tailored to endometriosis patients, with consideration of both global findings and regional feasibility.

Since endometriosis is a chronic condition, it is essential for patients to adopt dietary habits that enables them to cope with their symptoms and may also help reduce the disease severity. A study conducted in Australia reported that a change in dietary habit is one of the most popular self-management strategies for women with endometriosis, with 44% of all patients employing it (74). According to Mazza et al. (5), some of the most popular diet regimens chosen by endometriosis patients included gluten-free diet (15%), anti-inflammatory diet (8%), Mediterranean diet (7.1%), and ketogenic diet (4%). Moreover, participants in the study started to eat more fruits and vegetables (10%), cereals and legumes (6.6%), and fish (4.5%). They also cut back on dairy products (18.4%), soy-containing food (6.7%), and saturated fats (8%). Successful adherence to these dietary habit modifications requires the patients to be aware of the role of nutrition in endometriosis prevention and management, which is why health promotion and patient education for this matter are essential.

An Italian study found that 52% of endometriosis patients who adhered to a gluten-free diet for 12 months reported a significant reduction in pain severity (75). Similarly, an Austrian study investigated the benefits of Mediterranean diet for endometriosis and observed notable improvements in pain, dysmenorrhea, dyspareunia, dyschezia, and overall well-being (76). The Mediterranean diet has been associated with reduced homocysteine levels and increased serum concentrations of folate and vitamin B12, nutrients that are believed to attenuate inflammation and oxidative stress, both of which are implicated in the pathophysiology of endometriosis (55).

Overall, current evidence indicates that individuals with endometriosis may benefit from adopting a diet rich in antioxidants and anti-inflammatory nutrients while minimizing the intake of pro-inflammatory foods. The consumption of alcohol and trans fats, and saturated fats should be limited and patients with endometriosis are encouraged to replace them with PUFAs. Any dietary modifications should be made in consultation with a healthcare provider, and referral to a clinical nutritionist should be considered as part of a comprehensive management plan.

Establishing and maintaining a healthy dietary pattern is a long-term commitment that requires ongoing monitoring, individualised assessment, and sustained motivation.

Other lifestyle recommendations

In a study by Youseflu et al. (77), the impact of endometriosis on the sleep quality of women was investigated, with a focus on the potential influence of lifestyle factors. The study revealed that endometriosis adversely affects sleep quality (SQ), as patients report lower scores in subjective sleep quality, sleep latency, and sleep disturbance compared to healthy controls. Lifestyle modifications such as diet and physical activity play a crucial role in modifying SQ in endometriosis patients. The consumption of dairy products and nuts, which are rich in tryptophan, has been associated with better SQ due to their influence on melatonin production and regulation of inflammation. Regular physical activity enhancing melatonin secretion and improves SQ, although issues such as pain or abnormal menstrual cycles may decrease patients' tendency for exercise. In summary, counseling on diet strategies and promoting regular physical activity are recommended to improve SQ and overall well-being in endometriosis patients (77).

Another study reported that lifestyle interventions, particularly regular physical exercise, may exert protective benefits against the progression and severity of endometriosis symptoms (78). The discussion highlights the potential role of regular physical exercise in mitigating inflammatory conditions, including endometriosis. Despite evidence suggesting an association between endometriosis and inflammation, the role of exercise in managing the condition remains unclear due to a lack of well-designed, controlled clinical trials. While observational studies suggest a positive inverse relationship between exercise and endometriosis risk, these associations have yet to be confirmed through robust empirical evidence. In addition to physical activity, other lifestyle strategies such as stress reduction techniques and improved sleep hygiene may also contribute to better symptom management. However, further research is necessary to validate their effectiveness and to determine their specific impact on symptom burden and quality of life in patients with endometriosis (78).

This review is subject to several limitations. First, the findings presented were based on data from observational studies, case-control studies, and a limited number of randomized controlled trials, which may not necessarily reflect causal relationship. Furthermore, as the studies took place in different countries and used self-reported dietary instruments, there could be some extent of recall bias. Future research must therefore consider the effect of sociocultural and geographical diversity on dietary pattern and subsequent endometriosis risk.

Conclusion

The relationship between diet and endometriosis is complex, with certain foods either worsening or potentially protecting against the condition. Consumption of alcohol, red meat, processed foods, and excessive caffeine

may increase inflammation and hormonal imbalance, which may worsen endometriosis symptoms. On the other hand, nutrition such as antioxidants, B vitamins, vitamin D, calcium, omega-3 and omega-6 fatty acids, and fiber may help suppress inflammation and protect against the disease. Healthcare providers should promote a balanced diet rich in anti-inflammatory foods while limiting pro-inflammatory substances. A diet focused on fruits, vegetables, lean proteins, whole grains, and healthy fats is beneficial. Other lifestyle changes like exercise, stress management, and good sleep hygiene should also be part of treatment. Further research is required to elucidate the mechanisms by which dietary factors influence endometriosis and to establish evidence-based, individualized dietary guidelines for affected patients.

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Author's Contributions

R.M., E.C.Y., A.N.I., K.M.; Conceptualization and Methodology. R.M., B.W., A.H.; Project administration and Supervision. E.C.Y., A.N.I., K.M.; Formal analysis and Writing the original draft. K.S., A.K.H., G.P., M.M., B.W., A.H.; Data curation, Review, and Editing. All authors read and approved the final manuscript.

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